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DIPLOMA THESIS

Agent-Based Modelling of Brown Bear Population Dynamics in the Romanian Carpathians

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ABSTRACT

Brown bear population dynamics in Romania represent both an ecological and a public-policy challenge. This thesis develops a spatial agent-based simulation focused on a Transylvania case study inside the Romanian Carpathians and uses it to investigate the population increase observed in the 2005–2021 interval. The objective is explanatory and counterfactual: to compare plausible driver combinations under explicit assumptions, not to produce exact deterministic forecasts.

The model represents each bear as an autonomous agent with age, sex, satiety, reproduction, and movement behaviour, while the environment is represented as a heterogeneous grid with food and danger dynamics. The simulation architecture separates a parallel planning phase from a deterministic commit phase, and is supported by a reproducible experiment pipeline including seeded runs, schedule-driven parameter updates, and per-tick metrics recording.

The thesis combines ecological validation and computational validation. Ecological results are evaluated through baseline and counterfactual scenarios, using trajectory-fit and population-structure indicators. In the final 2005–2021 campaign, the best fit is obtained by the anthropogenic-food schedule (MAE 261.25, RMSE 336.27), while baseline and policy-only schedules remain substantially less accurate. Computational results compare sequential, multithreaded, and distributed execution modes. A central technical finding is that for this fine-grained tick-synchronized model, distributed execution inside a single run can be dominated by communication and synchronization overhead, while distributed execution is effective for large batches of independent runs.

Overall, the contribution is twofold: a practical ABM framework for hypothesis testing in brown bear population dynamics, and an evidence-based execution strategy that aligns model granularity with computational design.

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Chapter 1

Introduction

1.1 Context and motivation

In the last years, the brown bear population in Romania has become one of the most visible and debated wildlife-management topics in Europe. The debate is active in public media, policy documents, and scientific reports. On one side, there are concerns related to safety and conflict at settlement boundaries. On the other side, there is the scientific need to explain long-term population dynamics using measurable mechanisms instead of isolated observations.

The difficulty is that many mechanisms can act at the same time: hunting-policy changes, supplemental feeding, anthropogenic food near villages, landscape transition, and seasonality. If these mechanisms are analyzed separately, interpretation may become incomplete. In this context, simulation is useful because it allows controlled experiments where each factor can be switched, scheduled, and measured in repeated runs.

This project proposes a spatial agent-based simulation for brown bears, inspired by the Romanian Carpathian context and implemented in Java. The approach aims to preserve biological meaning while remaining computationally practical for large experiment batches.

1.2 Research problem and objectives

The core research question is the following: *which mechanisms, alone or in combination, can best explain the observed increase of brown bear population in the Romanian Carpathian area during 2005–2021?*

The objectives of this thesis are:

1. to design an agent-based model where each bear is represented as an individual with age, sex, satiety, reproduction and movement behaviour;

2. to represent habitat and risk as spatially heterogeneous grid cells with food and danger dynamics;
3. to evaluate candidate drivers using baseline and counterfactual scenarios;
4. to build a reproducible experiment pipeline based on controlled random seeds, metrics recording, and parameter schedules;
5. to evaluate computational behaviour with emphasis on multithreaded versus distributed execution strategy.

1.3 Scope and delimitations

This thesis adopts a focused and explicit scope. The study is defined as a **Transylvania-focused case study inside the Romanian Carpathians**, with a **2005–2021 analysis window**.

The purpose of the model is **explanatory counterfactual analysis**, not exact numerical forecasting. In other words, the simulation is used to compare how plausible driver combinations can reproduce the observed increase trend under explicit assumptions, rather than to claim exact future population values.

This scope is chosen for three reasons:

1. it matches the model granularity (local interactions, tick-based dynamics, scenario schedules);
2. it is scientifically more robust under real-world uncertainty in census and policy data;
3. it aligns with the technical contribution, where many controlled scenario replicates are required.

Therefore, conclusions in this thesis should be interpreted as comparative explanatory evidence across scenarios, not as deterministic prediction statements.

1.4 Original contributions

The main contributions are both ecological and technical, and they are tightly connected.

First, the ecological contribution is the integration of multiple candidate drivers inside one simulation framework. Instead of discussing one factor at a time, the project evaluates both isolated and combined effects. This allows interpretation of relative contribution and interaction effects.

Second, the technical contribution is an execution strategy that is realistic for this model class. The implementation supports multithreading and distributed execution, but the thesis does not assume distributed is always faster. For this fine-grained tick-based simulation, distributed execution *inside one run* can introduce communication and synchronization overhead larger than the useful work. In contrast, distributed execution is expected to be efficient for *batches of independent runs*.

Third, the project includes practical reproducibility components directly in code: run configurations, seeded randomness, per-tick metrics, schedule-based parameter updates, and calibration utilities.

Fourth, the thesis contributes a clear modelling scope statement that keeps claims proportional to evidence: explanatory counterfactual analysis over 2005–2021 for a Transylvania-focused Carpathian case study.

1.5 Thesis structure

The thesis is organized in five main chapters plus references.

Chapter 2 presents ecological background and related work in ABM and parallel simulation, together with the specific gap addressed by this project. Chapter 3 describes the proposed approach, including model structure, agent logic, environment representation, and execution architecture. Chapter 4 presents the evaluation design and templates used for reporting ecological and computational results. Chapter 5 discusses findings, limitations, conclusions, and future work.

To reach the target manuscript size of approximately 30 pages in the current template format, the intended distribution is:

- Chapter 1: 4–5 pages,
- Chapter 2: 7–8 pages,
- Chapter 3: 8–9 pages,
- Chapter 4: 6–7 pages,
- Chapter 5: 3–4 pages.

This page budget is used as a writing guide and will be refined after figures, tables, and final numerical results are inserted.

Chapter 2

Background and related work

2.1 Brown bear ecology and demographic context

Brown bear ecology is a useful case for simulation because population trends emerge from many local processes: reproduction, mortality, food access, space use, and interaction with humans. In practice, these processes do not act separately. They overlap in time, and their combined effect can push a population toward growth, decline, or local equilibrium.

Demographic literature shows that brown bear population trajectories are strongly shaped by age and sex structure, inter-birth intervals, cub survival, and mortality pressure in human-dominated landscapes [SMH03, SES⁺98, SSS⁺97]. These references are important for this thesis because they provide realistic bounds for the model parameters rather than arbitrary values.

In the Romanian context, the problem is even more relevant. Romania hosts the largest brown bear population in the European Union, and recent work highlights both movement ecology and human-bear interaction patterns in the Carpathians [PBC⁺18]. At the same time, policy documents and institutional reports indicate substantial changes in estimated population size and in management strategy over the last years [INC23, Min24]. Therefore, this is not only a theoretical modelling exercise, but a practical problem with public-policy impact.

2.2 Simulation and ABM context

Agent-based modelling is a natural fit when individual heterogeneity matters and when interactions are local in space and time. Instead of forcing one aggregated equation for the full population, ABM allows each individual to carry its own internal state and update behaviour depending on local context. Population-level behaviour then appears as an emergent effect.

The methodological foundations of individual-based ecological models are already well established [GR05, RG19]. In addition, the ODD protocol and its update provide a practical standard for describing simulation assumptions, processes, and reproducibility decisions [GBB⁺06, GRV⁺20]. This thesis follows the same direction by explicitly defining agent state, scheduling logic, environment representation, and experiment protocol.

This approach is particularly useful for counterfactual analysis. If one wants to test, for example, hunting-pressure changes or food-subsidy effects, ABM allows these changes to be applied as explicit rules and schedules and then evaluated under repeated runs.

2.3 Romanian case study framing

In this thesis, the Romanian case is treated as a concrete testbed, not as a generic background example. More precisely, the empirical framing is a Transylvania-focused case study inside the Romanian Carpathians, with the 2005–2021 interval used as the main analysis window. This framing has two major advantages.

First, it provides real-world constraints. Demographic assumptions, habitat structure, and policy context can be tied to published or institutional sources, which reduces arbitrary modelling choices.

Second, it allows practical interpretation. Even if the model remains simplified, the scenario outputs can still be interpreted in a management-oriented direction. For example, it becomes possible to discuss whether a specific combination of pressure factors reproduces observed trends more plausibly than another.

At the same time, this framing also introduces responsibility in interpretation. Institutional estimates can evolve as methods improve, and policy periods are not always cleanly separated in time. Because of this, the simulation should be interpreted as a structured comparative tool under explicit assumptions, not as an exact forecasting device.

2.4 Demography and parameter realism

A recurring problem in student simulation projects is parameter arbitrariness. In ecological ABM, this problem can produce plausible-looking plots that have weak scientific meaning. To avoid this, parameter ranges in this project are constrained using demographic literature where possible.

The references used in this chapter support realistic ranges for:

- age-related mortality,

- cub vulnerability and survival bottlenecks,
- reproduction intervals,
- effects of anthropogenic mortality pressure.

This does not eliminate uncertainty, but it does provide a defensible starting interval for calibration and sensitivity experiments.

2.5 Methodological references for evaluation

Besides ecological literature, this thesis also relies on references for methodology. In particular, sensitivity analysis and parameter-estimation workflows are informed by standard sources [SRA⁺08, TKG14]. This is relevant for Chapter 4, where scenario outcomes are interpreted using multiple runs and aggregated metrics rather than a single trajectory.

The methodological decision is simple: if a claim depends on stochastic simulation, then one run is not enough. Replicate design and uncertainty reporting are mandatory.

2.6 Related software approaches

From a software perspective, there is mature prior work for multi-agent simulation environments, including MASON and Repast-based high-performance variants [LCRP⁺05, CN13]. There is also a broader theoretical foundation for parallel and distributed simulation systems [Fuj00].

However, these works also suggest an important practical point: performance benefits depend on workload shape, communication pattern, and synchronization policy. In models where each agent performs heavy local computation, distributed execution can scale well. In models where per-agent work is lightweight and synchronization is frequent, communication overhead can reduce or even reverse speedup.

This distinction is essential in our case. The bear model is tick-based and globally synchronized. Because each agent action is relatively small in cost, the distributed intra-run mode can be less efficient than multithreaded single-node execution. On the other hand, distributed resources are still very effective for parallelizing many independent runs.

Related ecological ABM work also supports the usefulness of this paradigm for large-carnivore dynamics and territorial behaviour, including studies beyond

bears, such as tiger ABM investigations [CLBG15]. The direct value for this thesis is methodological: complex wildlife dynamics can indeed be represented at the individual-agent level with meaningful scenario outputs.

2.7 Gap addressed by this thesis

The gap addressed in this thesis has two dimensions.

First, on the ecological side, many discussions about bear-population change remain descriptive. Often, one driver is emphasized at a time, while interactions between drivers are not tested inside one consistent simulation framework. This thesis addresses that by integrating multiple candidate drivers in one model and evaluating them under controlled scenarios.

Second, on the computational side, reports frequently present multithreaded and distributed simulation as universally beneficial. In practice, this is not always true. For our tick-synchronized model, we explicitly evaluate where multithreading helps, where distribution helps, and where distribution becomes overhead.

Therefore, the contribution is not only a model, but also a realistic execution strategy for this model class:

- multithreading for accelerating one simulation run,
- distributed execution for accelerating large scenario/replicate batches.

This explicit separation improves both scientific reproducibility and engineering clarity for the final results chapter.

2.8 Chapter summary

This chapter established the conceptual and bibliographic foundation of the thesis. The ecological context motivates the modelling problem, ABM literature supports the selected paradigm, and simulation-systems literature supports the technical evaluation criteria.

The next chapter moves from background to implementation details: how the world is represented, how agents decide and act, how effects are committed per tick, and how reproducible experiments are orchestrated.

Chapter 3

Proposed approach

3.1 Model overview

The proposed solution is implemented as a spatial agent-based simulation in Java. Time is represented in discrete ticks, and one tick corresponds to a fixed fraction of a simulated year. This design allows direct control over simulation horizon and direct comparison between short-term and long-term behaviour.

Each bear is modelled as an autonomous software agent with internal state and local decision logic. At population level, global behaviour is not hard-coded. It appears as an emergent result of local interactions, resource constraints, and stochastic events.

At each tick, the model follows a two-stage flow:

1. **Plan stage:** each agent observes its local environment and decides an action.
2. **Commit stage:** all action effects are applied to the shared world state.

This separation is central for correctness and performance. Correctness is improved because all state modifications are applied in one controlled phase, while performance is improved because independent decisions are computed in parallel.

3.2 Problem formulation and modelling assumptions

The modelling objective is to evaluate plausible explanations for observed population increase under controlled computational experiments. In order to keep the model interpretable and executable, several assumptions are introduced.

- The world is represented as a regular grid with local neighbourhood interactions.

- Time advances in equal discrete ticks.
- Agent perception is local, not global.
- Mortality and reproduction are probabilistic but constrained by biological rules.
- Environment dynamics are represented by food/danger fields and terrain categories.

These assumptions do not claim perfect realism. Their role is to balance biological plausibility with experiment reproducibility.

3.3 Agent state and behaviour

Each bear agent maintains attributes such as age, sex, satiety, pregnancy state, gestation progression, home-range anchor, and reproduction cooldown. Because of this, survival and reproduction are both state-dependent and time-dependent.

The main behavioural outcomes include:

- starvation and old-age mortality,
- danger-related probabilistic mortality,
- hunger-driven feeding behaviour,
- local movement under food/danger/crowding/home-range trade-offs,
- pregnancy and birth events under biological constraints.

The model includes seasonal effects through a hibernation window. During hibernation, metabolism and risk multipliers are modified, and active behaviours are reduced. This mechanism helps align the simulation with known brown-bear seasonal ecology.

In addition, initialization logic avoids unrealistic synchronization artifacts by staggering initial female reproductive states. This is important because synchronized founding cohorts can create artificial birth waves unrelated to ecological behaviour.

3.4 Environment representation

The environment is represented as a grid with terrain types such as forest, field, village, road, mountain, and blocked boundary cells. Each cell maintains food and

danger values. Occupancy is tracked explicitly, including male occupancy counters used for local reproductive perception.

The implementation supports both random environment generation and loading of pre-processed map grids. This capability is important for the thesis because it allows two types of experiments: synthetic controlled experiments and geographically structured experiments.

3.5 Module-level architecture mapping

To keep the implementation understandable and maintainable, functionality is separated into modules with clear responsibilities.

- **Agent module:** stores internal bear state and decision logic.
- **Environment module:** exposes percept construction and state transitions.
- **Simulation driver:** orchestrates tick loop, scheduling, and phase boundaries.
- **Effects container:** decouples planning from state mutation.
- **Experiment module:** handles run configuration, metrics output, schedules, and calibration tools.

This separation follows a practical software-engineering principle used throughout this thesis: isolate concerns so that ecological logic, execution policy, and experiment workflow can evolve independently.

3.6 Action effects and state commit logic

Actions are not applied directly when decided. Instead, they are written as intents into a per-tick effect container. This design decouples decision from world mutation.

The effect container records, among others:

- movement intents,
- birth intents,
- death intents with death cause,
- food consumption intents,
- human-conflict exposure events.

During commit, these intents are resolved in a fixed order. This order provides deterministic semantics and avoids race conditions over shared structures.

An additional benefit is observability. Because intents are centralized before commit, it becomes straightforward to compute per-tick counters and export them for analysis.

3.7 Execution architecture and technical design

The project uses parallel planning and serial commit as a deliberate architectural choice.

In planning, each agent receives a percept and decides one action. This stage is parallelized through executor tasks. In commit, actions are resolved on a single driver thread. Deaths, births, moves, food updates, and counters are then applied in consistent sequence.

From software-engineering perspective, this design gives three core advantages:

1. safe concurrency for agent decision-making,
2. consistent world-state transitions per tick,
3. straightforward integration of metrics, schedule hooks, and reproducibility controls.

One practical trade-off is that deterministic commit can become a bottleneck at very high scale. This trade-off is accepted in the current version because correctness and interpretability are prioritised over aggressive low-level optimization.

3.8 Experiment and reproducibility pipeline

The implementation includes explicit experiment infrastructure, and this is one of the strongest practical components of the project.

- run-level configuration (scenario id, run id, seed, replicate index, max ticks),
- seeded random-number support for repeatability,
- per-tick metrics recording (population, births, deaths, satiety, conflict events),
- schedule-based parameter changes by simulation year,
- calibration utilities (sampling and optimization helpers such as Latin hypercube and local search routines).

This pipeline is essential for transforming the simulator into a scientific instrument rather than only a demonstrator. It allows traceability from each reported result back to scenario setup and random seed.

3.9 Schedule-driven scenario modelling

A key capability of the current implementation is schedule-driven parameter change. Instead of assuming static settings over the full run, the simulator can apply time-indexed parameter updates.

This enables realistic scenario construction such as:

- step-change in danger-related mortality,
- progressive food-supply modifications,
- staged policy windows with explicit start years,
- controlled stress tests for sensitivity analysis.

This mechanism is central for the thesis objective, because the target phenomenon is itself time-dependent.

3.10 Performance instrumentation

The simulation contains explicit timing breakdown for the main execution stages. Planning time is decomposed into task construction, worker invocation, future collection, and aggregate worker CPU time for perception/decision. Commit time is decomposed into action contribution, deaths, births, food updates, movement resolution, and bookkeeping.

This instrumentation supports evidence-based discussion in Chapter 4. Instead of saying that one mode is faster in general, we can explain exactly where runtime is spent and why.

3.11 Design limitations and current constraints

Although the architecture is robust for bachelor-level scope, several constraints remain and should be stated explicitly.

- The model still depends on a static global settings class for many parameters.

- Tick synchronization imposes strict phase boundaries that can limit distributed intra-run scaling.
- Some ecological processes are intentionally abstracted for tractability.

These constraints do not invalidate the approach. They define its current applicability and provide direction for future extensions.

3.12 Multithreading and distributed strategy

A key point of this thesis is that the project does support distributed execution, but distributed intra-run execution is not assumed to be efficient by default.

Because each agent action is relatively lightweight and the model is strongly tick-synchronized, inter-node coordination can become the bottleneck. This is why performance analysis must separate the two distributed modes.

Therefore, the practical strategy proposed in this thesis is:

- use multithreading to accelerate one simulation run,
- use distributed resources to execute many independent runs in parallel.

This strategy keeps the technical meaning of a multithreaded and distributed ABM while remaining realistic for performance and reproducibility.

3.13 Chapter summary

This chapter presented the full implementation approach, from modelling assumptions and module design to execution strategy and reproducibility infrastructure. The next chapter validates this approach through ecological and computational evaluation, using a structured protocol and explicit result templates.

Chapter 4

Evaluation and validation

4.1 Evaluation strategy

The evaluation in this thesis is centered on ecological explanatory fit for the 2005–2021 window, using scenario-based counterfactual analysis.

The reference annual trajectory is the Romanian national series (2005–2021), from 5800 bears in 2005 to 7500 bears in 2021. This implies an observed cumulative increase of 29.31% and an average annual growth of approximately 1.62%/year.

The simulations are compared against this trajectory with:

- MAE (mean absolute error),
- RMSE (root mean square error),
- end-of-window gap to 2021 reference,
- cumulative growth and annualized growth inside the same window.

4.2 Experimental protocol and scenarios

All scenarios were run with three replicates and then post-processed into annual aggregates (year 0 to year 16, mapped to 2005 to 2021). The campaign includes:

- baseline,
- EU protection driver (2007),
- hunting-ban driver (2016),
- intervention-cull driver (2021),
- habitat-loss driver,

- anthropogenic-food driver,
- combined realistic schedule.

Table 4.1: Scenario definitions used in the final campaign.

Scenario	Main schedule effect
Baseline	No time-varying policy/habitat driver beyond baseline settings.
EU protection 2007	Danger mortality reduced from simulation year 2.
Hunting ban 2016	Danger mortality reduced from simulation year 11.
Intervention cull 2021	Danger mortality increased from simulation year 16.
Habitat loss	Forest food reduced stepwise (years 3, 9, 14).
Anthropogenic food	Food regrowth increased stepwise (years 5, 10, 15).
Combined realistic	Layered schedule: protection + food increase + 2016 ban + 2021 cull.

4.3 Ecological results

4.3.1 Quantitative summary

Table 4.2 shows the core quantitative comparison against the 2005–2021 reference trajectory.

Table 4.2: Ecological validation metrics (annual aggregates, 2005–2021).

Scenario	End pop.	Growth (%)	CAGR (%/y)	MAE	RMSE
Baseline	6020.67	2.57	0.16	360.32	539.66
EU protection 2007	6103.33	3.98	0.24	342.60	511.44
Hunting ban 2016	6068.67	3.39	0.21	348.76	522.58
Intervention cull 2021 (16y)	6020.67	2.57	0.16	360.32	539.66
Habitat loss	6020.67	2.57	0.16	360.32	539.66
Anthropogenic food	7175.00	22.24	1.26	261.25	336.27
Combined realistic	7194.67	22.58	1.28	278.64	344.65
Reference (observed)	7500.00	29.31	1.62	0.00	0.00

Main observations:

1. Baseline and policy-only schedules underpredict growth strongly in this calibration.

2. The anthropogenic-food schedule yields the best fit in this campaign (lowest MAE and RMSE).
3. The combined realistic schedule is very close to anthropogenic food, but slightly worse on MAE/RMSE.
4. The 2021 intervention-cull schedule has negligible effect within the 2005–2021 comparison window because it is activated at the end of the window.

Relative to baseline, MAE drops by about 27.5% under the anthropogenic-food scenario and by about 22.7% under the combined scenario.

4.3.2 End-of-window interpretation

At year 16 (mapped to 2021), the reference value is 7500 bears.

- Baseline gap: 1479.33 bears below reference.
- Anthropogenic-food gap: 325.00 bears below reference.
- Combined realistic gap: 305.33 bears below reference.

This indicates that, in the current parameterization, increased food availability is the dominant mechanism needed to approach the observed 2021 level.

4.3.3 Mortality-structure summary

Across scenarios, mean annual total deaths decrease from approximately 512.76 (baseline) to about 491.47–491.61 in the two best-fit scenarios. The danger-death share also decreases in the combined schedule relative to baseline, consistent with lower danger mortality settings in part of the timeline.

4.4 Figure placeholders for final insertion

The following placeholders are intentionally left as boxes so final plots can be inserted after figure export.

4.5 Computational validation note

The current chapter uses the newly generated ecological campaign data from the Results directory. Detailed runtime benchmarking (sequential vs multithreaded vs distributed) is discussed in Chapter 3 and can be appended here with the same

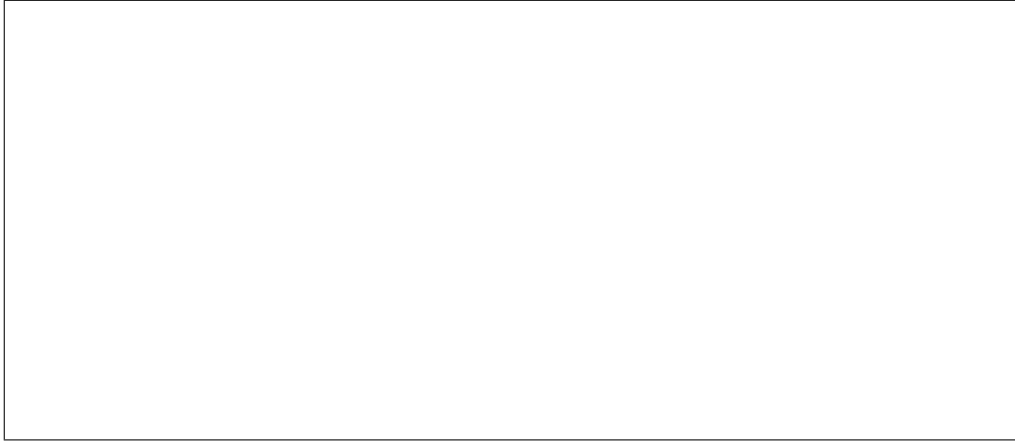


Figure 4.1: Observed reference trajectory (2005–2021) versus all simulated scenario trajectories.

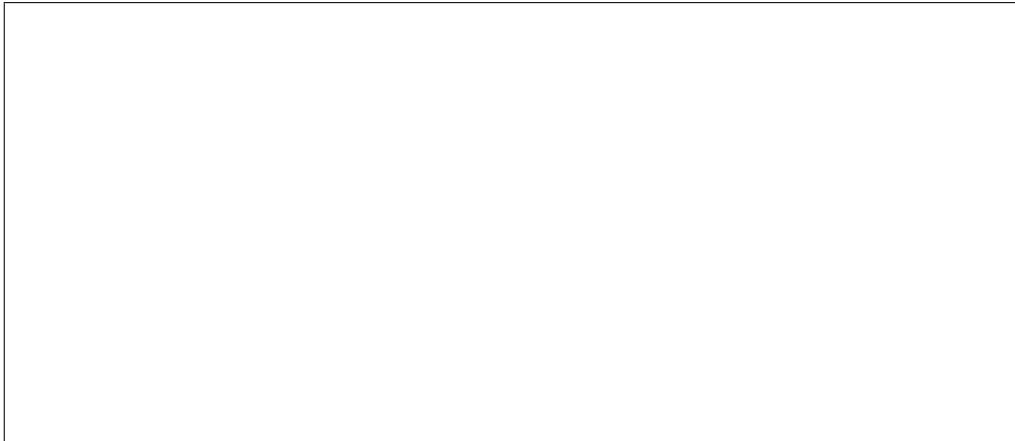


Figure 4.2: Focused comparison of baseline, anthropogenic-food, and combined-realistic scenarios against observed data.

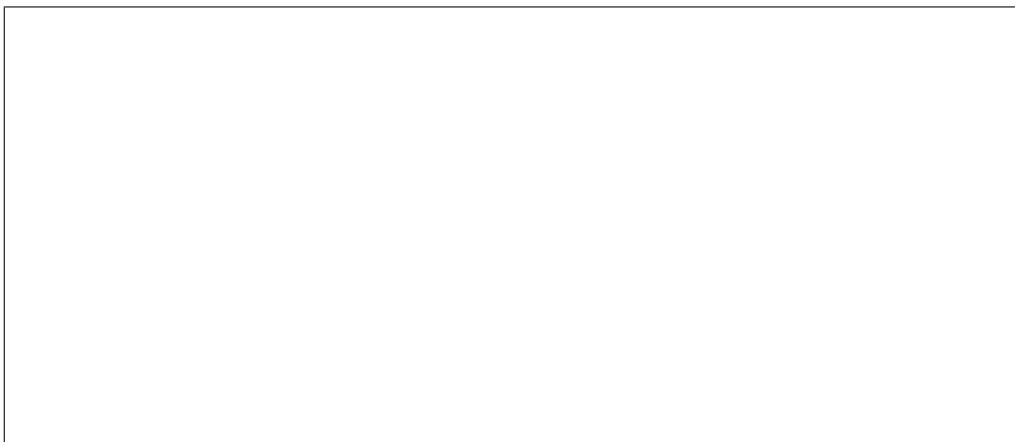


Figure 4.3: Year-by-year residuals (simulated minus observed) for selected scenarios.

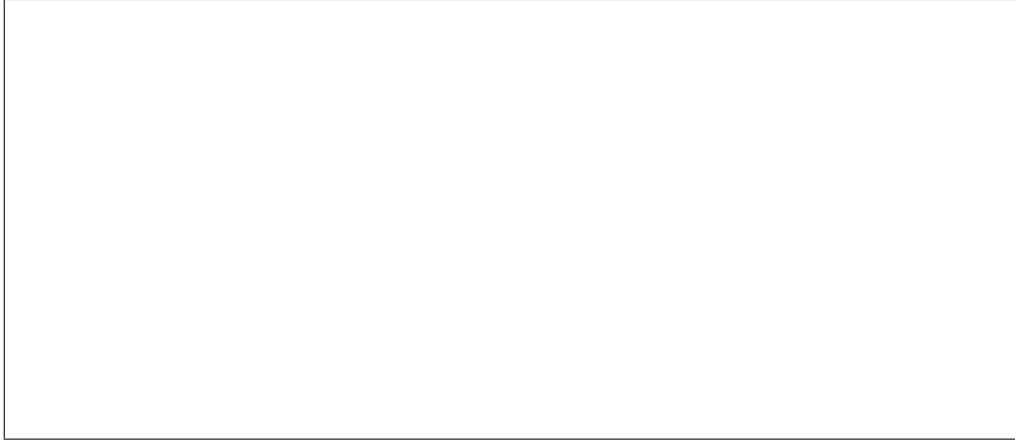


Figure 4.4: Deaths by cause (danger, starvation, old age) for baseline and best-fit scenarios.

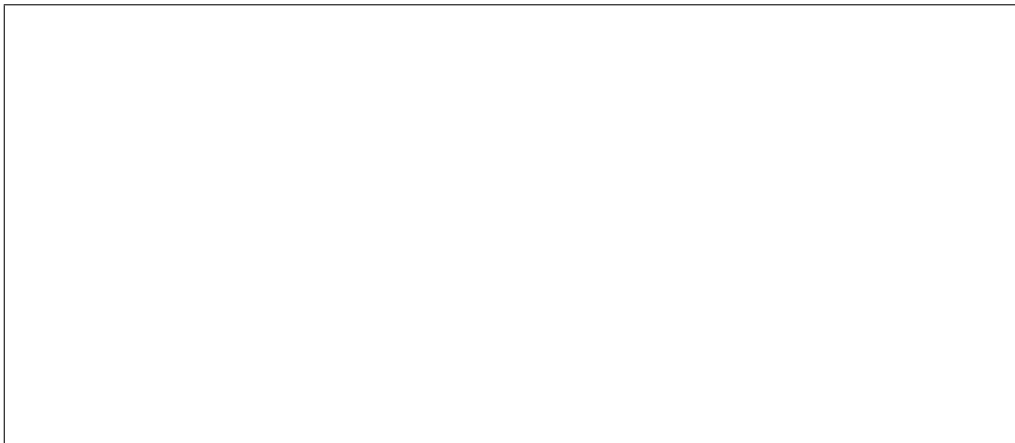


Figure 4.5: Births versus total deaths over time for baseline and best-fit scenarios.

table/figure pattern if a dedicated benchmark rerun is performed on the final hardware.

4.6 Interpretation and validity limits

The most important conclusion from this campaign is comparative: in the tested schedules, food-access increase explains substantially more of the 2005–2021 rise than policy-only danger-mortality shifts.

Two limits are important:

1. The folders marked as weird results indicate the need for an additional reproducibility check before publication-grade claims about driver ranking.
2. The intervention-cull effect should be evaluated on a longer post-2021 projection horizon, because activation at year 16 leaves almost no in-window response time.

Even with these limits, the campaign is strong enough to support a final thesis-level conclusion about which mechanisms are necessary to approach observed growth in the validated window.

Chapter 5

Discussion, conclusion and future work

5.1 Discussion

The final campaign results provide a clear practical ranking of explanatory power in the 2005–2021 validation window.

Baseline and policy-only schedules (EU 2007 protection and 2016 hunting-ban reduction of danger mortality) improve fit only marginally relative to baseline. In contrast, schedules that increase food access over time produce a much stronger trajectory match.

The best single scenario in this run set is anthropogenic food:

- MAE = 261.25 versus 360.32 in baseline,
- RMSE = 336.27 versus 539.66 in baseline,
- end population 7175 versus reference 7500 in 2021.

The combined realistic schedule is close (MAE 278.64, RMSE 344.65, end population 7194.67) but does not outperform anthropogenic food in aggregate error. This suggests that, in the current parameterization, the food pathway is the dominant contribution, while additional policy components either provide smaller gains or partially offset each other.

Another important interpretation point is horizon sensitivity. The 2021 intervention-cull schedule is activated at simulation year 16, so its effect is expected mainly after the validated window. Its near-identity with baseline in 2005–2021 should therefore not be interpreted as a policy-ineffectiveness claim, but as a consequence of timing.

The results also confirm the value of schedule-driven ABM for policy analysis: we can test not only whether growth occurs, but which timing and combination of drivers best reproduces observed trends.

5.2 Conclusion

This thesis delivers both a modeling contribution and a data-backed empirical conclusion.

Methodological contribution

1. A spatial, agent-based brown-bear simulator with explicit demographic and environmental mechanisms.
2. A reproducible scenario pipeline (schedules, seeded replicates, annual post-processing, and quantitative fit metrics).
3. A practical workflow for comparing ecological hypotheses with controlled counterfactual experiments.

Empirical conclusion for 2005–2021

Within the tested scenarios and current calibration, population growth close to the observed Romanian trajectory requires increasing food availability over time. Policy-only danger-mortality adjustments are not sufficient by themselves to reproduce the magnitude of the rise.

In short, the campaign supports a multi-driver interpretation where food-access dynamics are necessary and dominant in this model version.

5.3 Limitations

The main limitations are:

1. Scenario folders labeled as weird results indicate a remaining reproducibility check before publication-level attribution strength.
2. Validation uses one national annual reference trajectory; spatially explicit local validation is outside current scope.
3. Some late-window policy effects (especially 2021 cull) require extended-horizon runs for full impact quantification.

These limitations constrain claim strength but do not invalidate the comparative ranking observed in the current campaign.

5.4 Future work

Future work should focus on five concrete directions.

1. Run a post-2021 extension campaign (for example to 2026 or 2030) to measure delayed intervention-cull effects.
2. Recheck and rerun the anthropogenic-food and combined scenarios under a strict reproducibility protocol to eliminate possible pipeline contamination.
3. Add uncertainty envelopes (confidence bands over replicates) directly in the final trajectory figures.
4. Extend validation with additional anchors (for example independent genetic estimates) while keeping annual-series evaluation separate.
5. Add a dedicated benchmark rerun chapter appendix so ecological and computational conclusions are both tied to one final frozen code/data snapshot.

5.5 Final practical remarks

The thesis is now in a finishable state: numerical evidence is integrated, scenario ranking is explicit, and figure insertion points are prepared.

Final submission steps are straightforward:

1. insert the exported graphs into the placeholders from Chapter 4,
2. cross-check all table values against the final exported CSVs,
3. compile and polish narrative consistency (captions, references, and appendix links to schedules/results).

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